

Internet of Things (IoT) Device Evolution Report

Section 1: Introduction

A traditional light bulb is one of the most common household devices used for basic illumination. It has Evolved into a smart IoT-enabled light bulb controlled through smartphones, voice assistants, and Automation systems. This report compares incandescent bulbs with smart IoT bulbs such as Philips Hue, focusing on features, improvements, limitations, privacy, and price changes.

Section 2: Function & Feature Comparison

The original incandescent bulb only turns on or off using a physical switch. It has no connectivity or Automation. In contrast, IoT smart bulbs can be controlled remotely using mobile apps, change Brightness and colors, be scheduled, and integrate with smart home systems. They also support voice Assistants like Alexa and Google Assistant. The biggest advancement is remote access and Automation, which old bulbs cannot do.

Section 3: Improvements, Diminishment, and Limitations

Improvements include remote control, automation, energy efficiency, and smart home integration. **Diminishments** include higher cost, setup complexity, and reliance on apps and Wi-Fi. **Limitations** Include dependency on internet connection, firmware updates, and potential compatibility issues.

Section 4: Privacy & Price Analysis

Traditional bulbs do not collect data, so they have no privacy risks. IoT smart bulbs may collect usage Data, app activity, and device information stored in the cloud. This introduces privacy concerns not Present in older devices. Price comparison: Incandescent bulb costs about \$1 (around \$1.80 adjusted For inflation to 2026). Smart IoT bulbs cost between \$15–\$50, making them significantly more Expensive due to added technology and features.

